Master Mathematics
2026 - 2027

(Mathematics) compulsory courses	54-60 EC									
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*		
Current Issues in Mathematics	new	6	3	3					*to replace XM_0095 Introduction to Master Mathematics*	*level 400, pass/fail, no exam, HC, only open to M MAT, suggestions teacher: Stephanie, Raffaella*
Master Project Mathematics	X_400355	36								
									in 25-26, 2 or 3; in 24-25, 1; earlier 1 or 2 -> STEPHANIE IS OK WITH 2 OR 3, DOUBLE-CHECK WITH OLC, smth for the OER	
(Mathematics) seminars	**constrained choice: choose 2 or 3 (12-18 EC)**									
Sequential Monte Carlo Methods	XM_0182	6	3	3					*aimed at: Data and Computation*	*p4+5 -> p1+2, Frank*
Data Assimilation	new	6	3	3					*aimed at: Computation and Data*	*Svetlana, level 400, no exam, HC, only open to M MAT*
Geometry of Statistical Models	new	6				3	3		*aimed at: both but more Data than Computation*	*Rikkert, level 400, no exam, HC, only open to M MAT*
Covering Spaces in Topology and Algebra	new	6				3	3		*aimed at: Computation*	*Inbar, level 400, no exam, HC, only open to M MAT*
Uncertainty Quantification (new in 25-26, Svetlana, p1+2), aanbodperiode verwijderen										
p-Adic Numbers and Their Applications (new in 25-26, Rob, p1+2), aanbodperiode verwijderen										
Network Dynamics, Symmertry and Synchrony (new in 25-26, Chris, p4+5), aanbodperiode verwijderen										

Mathematics and Data study path

no change from "and" to "of" per OLC advice

(Mathematics and Data) core electives	18 EC								
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	
Mathematical Foundations of Machine Learning	new	6	6						*to rename XM_0010 Advanced Machine Learning, compulsory from 27-28 on*
Statistical Models	X_400418	6	3	3					
Mathematical Optimization	XM_0051	6		6					

(Mathematics and Data) focus electives	choose at least 2 (>=12 EC)								
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	
Applied Stochastic Modelling	X_400392	6	3	3					
Dynamic Programming and Reinforcement Learning	XM_0093	6		6					
Large Language Models from scratch	new	6				6			*new course in MSc BA, follow-up on MFML+DPRL*
Optimization of Business Processes	X_400422	6				3	3		
Statistical Learning with High-Dimensional Data	new	6				3	3		*to rename X_405113 Statistics for High-Dimensional Data, alt, yes in 26-27*
Asymptotic Statistics	XMM_400323	8							
Measure Theoretic Probability	XMM_400244	8							
Stochastic Processes	XMM_400339	8							

(Mathematics and Data) suggested electives	<=36EC								*run these lists by 'path coordinators':*
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	*Data Sandjai and Frank, Bio BobP, AG Sander, ADS BobR and Daniele*
former DSO/PS track suggested electives, from 25-26, 24-25:									*done*
matches department's research profile:									
Performance of Networked Systems	ours								
Stochastic Processes for Finance	ours								
Continous Optimization	Mastermath								
Scheduling -> NEW NAME Scheduling Theory and Algorithms	Mastermath								
Queueing Theory	Mastermath								
Machine Learning Theory	Mastermath								
~~Stochastic Integration~~ NOT given in 26-27	Mastermath								
~~Statistical Learning~~ NOT given in 26-27	Mastermath								
Time Series	Mastermath								
~~Mathematical Biology~~ NOT given in 26-27	Mastermath								
Forensic Probability and Statistics	Mastermath								
High-dimensional Probability ADDED to suggestions in 26-27	Mastermath								
extra:									
Applied Analysis: Financial Mathematics (MSc)	ours								
~~Semidefinite Optimization~~ NOT given in 26-27	Mastermath								
Discrete Optimization	Mastermath								
~~Advanced Discrete Optimization~~ DROPPED from suggestions in 26	Mastermath								
~~Advanced Linear Programming~~ NOT given in 26-27	Mastermath								
~~Multivariate Statistics~~ NOT given in 26-27	Mastermath								
Causality	Mastermath								

Mathematics and Computation study path

(Mathematics and Computation) core electives	6 EC							
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
Mathematical Foundations of Machine Learning	new	6	6					

(Mathematics and Computation) focus electives	choose at least 2: >=16 EC								
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	
Algebraic Topology I	XMM_400600	8							
Dynamical Systems	XMM_400429	8							
Elliptic Curves	XMM_400505	8							
Formal Methods in Mathematics	XMM_0066	8							
Mathematical Biology *alt with MathNeuro, not in 26-27*	XMM_400504	8							*alt with MathNeuro, not in 26-27*
Mathematical Neuroscience	XMM_0068	8							*alt with MathBio, yes in 26-27*
Partial Differential Equations	XMM_400330	8							
Topological Data Analysis *alt, not in 26-27*	XMM_0036	8							*alt, not in 26-27*

(Mathematics and Computation) suggested electives in Biomedical Mathematics								<=44EC	
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	
former 30EC Bio package (VU but not ours):									*done*
From Molecule to Mind									
Fundamentals of Bioinformatics									
Introduction to Systems Biology									
Algorithms in Sequence Analysis									
~~From Protein to Cell~~ level 300, we (have to) drop in 26-27									
Mechanics and Thermodynamics in the Cell									
~~Metabolomics~~ level 300, we (have to) drop in 26-27									
Basic Models of Biological Networks									
Brain Imaging (for AI)									
Structural Bioinformatics									
Bioinformatics for Translational Medicine									
Tracer Kinetic Modelling									
former Bio track suggested electives, from 25-26, 24-25:									
matches department's research profile:									
Statistics for High-Dimensional Data -> NEW NAME									
Statistical Learning with High-Dimensional Data	ours								
Measure Theoretic Probability	Mastermath								
Numerical Methods for PDEs	Mastermath								
~~Bifurcation Theory~~ NOT given in 26-27	Mastermath								
Machine Learning Theory	Mastermath								
Stochastic Processes	Mastermath								
extra:									
Functional Analysis	Mastermath								
Inverse Problems in Imaging	Mastermath								

(Mathematics and Computation) suggested electives in Algebra, Geometry and Number Theory								<=44EC	
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*	
former AG constrained choice (also matches department's research):									*done*
Algebraic Number Theory	Mastermath								
Algebraic Geometry I	Mastermath								
Differential Geometry	Mastermath								
former AG track suggested electives, from 25-26, 24-25:									
matches department's research profile:									
Analytic Number Theory	Mastermath								
Modern Cryptography	Mastermath								
Algebraic Topology II	Mastermath								
Algebrac Geometry II	Mastermath								
Advanced Algebraic Geometry upgraded to 'matches department's research'	Mastermath								
Symplectic Geometry	Mastermath								
extra:									
~~Riemann Surfaces~~ NOT given in 26-27	Mastermath								
Commutative Algebra	Mastermath								
Selected Areas in Cryptology	Mastermath								
Coding Theory	Mastermath								

(Mathematics and Computation) suggested electives in Analysis and Dynamical Systems								<=44EC
Course Name	*Code*	*EC*	*P1*	*P2*	*P3*	*P4*	*P5*	*P6*
former ADS compulsory: (extra)								
Functional Analysis	Mastermath							
former ADS track suggested electives, from 25-26, 24-25:								
matches department's research profile:								
~~Optimal Transport~~ NOT given in 26-27	Mastermath							
Calculus of Variations	Mastermath							
Numerical Methods for PDEs	Mastermath							
Numerical Methods for SDEs upgraded to 'matches department's research' (Svetlana's, Frank's)	Mastermath							
Differenital Geometry	Mastermath							
Symplectic Geometry	Mastermath							
~~Bifurcation Theory~~ NOT given in 26-27	Mastermath							
Non-lnear PDEs	Mastermath							
Interacting Particle Systems upgraded to 'matches department's research' (Daniele's, Chris')	Mastermath							
extra:								
Applied Analysis: Financial Mathematics (MSc)	ours							
Inverse Problems in Imaging (we do inverse problems, but not imaging)	Mastermath							
~~Riemann Surfaces~~ NOT given in 26-27	Mastermath							
~~Group Theory and Dynamical Systems~~ DROPPED from suggestions for 26-27: too niche and not so relevant for the track, nor matches department's research	Mastermath							